Notes: Brown, Harlow, and Starks (JF, 1996)

Of Tournaments and Temptations: An Analysis of Managerial Incentives in the Mutual Fund Industry

Contents

1 Big picture

- **Question.** Does the tournament structure of the mutual fund industry create incentives for poorly performing managers to change portfolio risk differently from better performing managers?
- **Setting.** The paper studies 334 growth-oriented mutual funds followed by Morningstar. The raw return history runs from 1976 to 1991, and the core tournament tests are built from twelve annual tournaments spanning 1980 to 1991.
- **Core mechanism.** Mutual fund compensation is linked indirectly to relative performance because better-ranked funds attract disproportionately larger inflows of new assets. If payoff is convex in rank, interim losers should have more incentive to gamble for recovery.
- **Main prediction.** Managers who are interim *losers* should raise risk by more than interim *winners* over the remainder of the year.
- **Second prediction.** The effect should be stronger for newer and smaller funds, because these funds have more to gain from dramatic relative-performance reversals and may face fewer organizational or clientele constraints.
- **Empirical design.** For each year, the authors classify funds as winners or losers based on cumulative returns through month M and compare this classification to a post- versus pre-assessment volatility ratio. The evidence is summarized with 2×2 contingency tables and chi-square tests, and then reinforced with logistic regressions.
- **Main baseline result.** Interim losers are more likely than interim winners to end up in the high-risk-adjustment group. The effect is strongest when the interim ranking is taken in July and remains after excluding December returns.
- **Time-pattern result.** The tournament effect is weak in the early 1980s and becomes much stronger in 1986–1991, consistent with rising investor attention and rapid growth in the mutual fund industry.
- **Cross-sectional result.** The pattern is more pronounced for newer funds and, after normalization, also stronger for smaller funds.
- **Mechanism result.** Simulated unmanaged stock portfolios do not reproduce the same winner-loser asymmetry, suggesting that the pattern reflects active managerial choices rather than passive changes in stock volatility.
- **Robustness result.** The findings survive exclusion of December returns, stratification by ex ante total risk and beta, and logistic controls for cumulative past performance.
- **Economic takeaway.** Relative-performance tournaments can induce managers with bad interim results to alter portfolio risk in ways that may be privately rational but not aligned with investor preferences.

2 Data and measurement (key objects)

2.1 Why growth funds are the relevant laboratory

- The paper restricts attention to **growth-oriented mutual funds**.
- The authors give two reasons. First, growth funds are the most visible and widely ranked segment of the industry, so the tournament idea should be especially relevant for them.
- Second, growth funds have a high ex ante tendency toward taking risky positions, which makes them a natural place to look for strategic risk shifting.

Interpretation. The paper is not about the entire mutual fund universe. It is about the part of the industry where relative rankings and risk-taking incentives should be most salient.

2.2 Sample construction

- The data are monthly returns from Morningstar Mutual Fund Services.
- For each fund, returns are available from the later of either 1976 or the fund's inception, through the end of 1991.
- A fund enters a given yearly tournament only if it has return data for the *entire* calendar year.
- Table I reports annual descriptive statistics for the 1976–1991 sample. The number of growth funds rises from 109 in 1976 to 334 in 1991, and the aggregate dollars invested rise from about $8.93 billion to $134.25 billion.

Interpretation. The sample construction mirrors how published annual rankings are typically consumed: managers are effectively compared over a common calendar year.

2.3 Annual tournament structure

The paper assumes that the mutual fund tournament is run on an annual cycle. For each fund j and year y, interim performance through month M is measured as

$$RTN_{j,y}^{(M)} = \left[\prod_{m=1}^{M} (1 + r_{jmy})\right] - 1. \tag{1}$$

Here r_{jmy} is fund j's monthly return in month m of year y.

The paper allows the interim assessment month to vary from April through August:

$$M \in \{4, 5, 6, 7, 8\}. \tag{2}$$

Funds are then classified in two ways:

- **Median split:** above-median RTN are winners, below-median RTN are losers.
- **Quartile split:** top quartile are winners, bottom quartile are losers.

Interpretation. The paper does not rely on one arbitrary definition of "winner" and "loser." It checks whether the tournament logic survives across both broad and extreme classifications.

2.4 Risk adjustment ratio

The core risk measure is the fund's **risk adjustment ratio**:

$$RAR_{j,y}^{(M)} = \frac{\text{SD}\left(r_{jmy} : m = M+1, \ldots, 12\right)}{\text{SD}\left(r_{jmy} : m = 1, \ldots, M\right)}. \tag{3}$$

This is the ratio of post-assessment volatility to pre-assessment volatility.

- If $RAR > 1$, risk increases after the interim ranking.
- If $RAR < 1$, risk decreases after the interim ranking.

The paper computes RAR both:

- including all months through December, and
- excluding December returns, to reduce contamination from possible year-end window dressing.

Interpretation. This is the central object. The paper is not asking whether losers are absolutely riskier than winners. It is asking whether losers *change* risk more after learning that they are behind.

2.5 2×2 classification design

Each annual observation is placed into one of four cells:

Loser / low RAR	Loser / high RAR
Winner / low RAR	Winner / high RAR

The null hypothesis is that the return ranking and the risk-adjustment ranking are independent, so each cell should contain about 25% of observations. The alternative predicted by the tournament model is that

$$\Pr(\text{Loser, high } RAR) \text{ and } \Pr(\text{Winner, low } RAR) \tag{4}$$

should both be larger than under independence.

Interpretation. The contingency-table design is a clean nonparametric way to test whether bad interim performance and post-assessment risk increases line up more often than random chance would imply.

2.6 Additional objects used in the robustness tests

1. Cumulative past performance. For $n \in \{2, 4\}$ years of history, cumulative lagged performance is defined as

$$CUMRTN_{j,y}^{(n)} = \prod_{t=y-n}^{y-1} \prod_{m=1}^{12} (1 + r_{jmt}). \tag{5}$$

This variable captures whether a fund has been a repeated winner or loser before the current tournament year.

2. Control portfolios. The paper forms 250 simulated unmanaged portfolios of 75 stocks and another 250 of 150 stocks from CRSP data. These portfolios are fixed once formed and are used to assess whether the winner-loser risk pattern could arise passively at the asset level.

3. Volatility subtournaments. In 1986–1991, funds are sorted by prior-two-year total volatility and by beta into high, middle, and low volatility groups. The tournament test is then repeated inside each subgroup.

4. Fund demographics. The paper distinguishes:

- **Entrenched funds:** the 109 growth funds present since 1976.
- **New funds:** funds entering later in the sample.
- **Small versus large funds:** below versus above the year-specific median asset value.

Interpretation. These extra objects are all aimed at the same credibility question: is the basic loser-high-risk pattern really about incentives, or is it an artifact of style, composition, or passive volatility dynamics?

2.7 Descriptive facts from Table I

- The number of sample funds rises from 109 in 1976 to 334 in 1991.
- Aggregate assets under management rise from \$8.93 billion to \$134.25 billion.
- Median annual fund returns fluctuate widely, from −4.89% in 1990 to 35.32% in 1980 and 34.31% in 1991.
- Median annual standard deviations also vary sharply, with especially high volatility in 1987.

Interpretation. The wide dispersion in annual returns and volatility is important because it makes interim rankings economically meaningful: in some years a manager needs an enormous second-half rebound to catch up, while in others only a modest improvement is needed.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and central hypothesis

The paper's central prediction is written as

$$\left(\frac{\sigma_{2,L}}{\sigma_{1,L}}\right) > \left(\frac{\sigma_{2,W}}{\sigma_{1,W}}\right), \tag{6}$$

where L denotes interim losers, W denotes interim winners, and σ_1 and σ_2 are portfolio risk levels in the first and second subperiods of the year.

Interpretation. Interim losers are the funds with the strongest incentive to gamble for recovery. Interim winners may also change risk, but they should do so by less.

3.2 Why the mechanism is tournament-like

The underlying argument is a convex payoff story:

- Funds with high relative returns attract disproportionately larger inflows.
- Poor performance does not trigger equally large outflows.
- Since manager compensation is tied to assets under management, a manager's objective resembles a call option on relative rank.

Interpretation. This is why the model predicts stronger risk-taking by losers. They have little to lose from staying behind and a lot to gain from a successful comeback.

3.3 Baseline contingency-table test

For each tournament year and each assessment month M, the authors compute RTN and RAR, classify each fund into winner/loser and high/low RAR groups, and test whether the cell frequencies differ from the equal-frequency null.

Let the four cell shares be denoted by

$$\pi_{LL},\ \pi_{LH},\ \pi_{WL},\ \pi_{WH}, \tag{7}$$

where the first index refers to loser/winner and the second to low/high RAR. Under independence,

$$\pi_{LL} = \pi_{LH} = \pi_{WL} = \pi_{WH} = 0.25. \tag{8}$$

The tournament prediction instead requires

$$\pi_{LH} > 0.25 \qquad \text{and} \qquad \pi_{WL} > 0.25. \tag{9}$$

Interpretation. This specification is deliberately simple. It does not assume linearity. It only asks whether interim losers are disproportionately overrepresented in the high-risk-adjustment category.

3.4 Assessment-date choice and temporal partitions

The paper varies the interim assessment month M from 4 to 8 and studies:

- the full 1980–1991 period,
- two six-year subperiods,
- four three-year subperiods,
- and separate annual tournaments.

Interpretation. This is a way of checking both timing and evolution. If the mechanism is real, it should not depend on one exact month or one specific subperiod, although the strength of the pattern can vary.

3.5 Logistic regression with cumulative performance

To examine whether repeated past success or failure matters in addition to the current interim rank, the paper estimates logistic models of the form

$$\Pr(DVOLRAT_{jy} = 1) = \Lambda\Big(\alpha + \beta_1 InterimPerf_{jy} + \beta_2 CumPerf_{jy}^{(2)} + \beta_3 CumPerf_{jy}^{(4)}\Big), \quad (10)$$

where $\Lambda(\cdot)$ is the logistic cdf.

The dependent variable is an indicator for falling on one side of the RAR ranking, and the coefficients are interpreted as showing whether current and cumulative winner status predicts a lower-risk-adjustment outcome for winners, equivalently a greater tendency of losers to end up in the high-RAR group.

Interpretation. The logit is the paper's more parametric confirmation that repeated loser status matters, not just the current midyear setback.

3.6 Active versus passive risk changes

The paper's simulation exercise uses unmanaged control portfolios to ask whether the measured pattern could arise even if nobody actively altered portfolio composition. If the actual-fund frequencies differ systematically from simulated unmanaged portfolios, then the evidence is more naturally read as *managerial action*.

Interpretation. This is a very important design feature. Without it, one could always say that loser funds simply hold stocks whose volatility rises mechanically after they fall.

3.7 Volatility subtournaments

The last major specification checks whether the result survives when funds are only compared to other funds with similar ex ante total volatility or beta.

Interpretation. This matters because investors may compare "high-volatility growth funds" mostly to other high-volatility growth funds. The paper shows the result does not depend on pooling all growth funds into one crude tournament.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

There is no formal instrument in the paper. Identification comes from the timing structure of the annual tournament:

- interim performance is measured through month M,
- risk adjustment is measured after month M,

- and the test asks whether the latter depends systematically on the former.

Interpretation. Since the winner-loser classification is fixed before the post-assessment volatility is measured, the basic direction of causality is built into the research design.

4.2 Why reverse causality is relatively weak

A fund cannot become a loser in July *because* it raises risk in August. The rank variable is predetermined with respect to the post-ranking subperiod. This does not solve every omitted-variable concern, but it does rule out the simplest mechanical reverse-causality story.

4.3 Normalization across years

The paper aggregates observations across many annual tournaments only after normalizing the ranking variables within each year. This matters because a 2% interim return can represent a winner in one year and a loser in another.

Interpretation. The normalization makes the pooled evidence comparable across market environments with very different return and volatility levels.

4.4 Window dressing

A natural concern is that year-end reshuffling of portfolios for reporting or cosmetic reasons could mechanically alter measured risk. The paper addresses this by recomputing RAR excluding December returns.

Interpretation. The main winner-loser pattern barely changes when December is removed, so the result is not just a December-window-dressing artifact.

4.5 Common market volatility shocks

Another concern is that losers and winners may simply have different betas, so aggregate volatility changes could generate different RAR values even without any incentive effect. The volatility-subtournament exercise addresses this directly by repeating the test within groups with similar ex ante total risk or systematic risk.

Interpretation. The result survives inside beta and volatility subgroups, which is strong evidence against the pure aggregate-volatility explanation.

4.6 Active management versus passive asset-level volatility

The simulated CRSP control portfolios are crucial. If actual mutual funds behaved just like unmanaged stock baskets, then the tournament interpretation would be much weaker. Instead, actual-fund frequencies are more skewed in the predicted direction than the control frequencies.

Interpretation. This is probably the paper's strongest argument that the effect reflects active management rather than passive drift in the underlying securities.

4.7 Alternative peer groups and style misclassification

Because investors may not evaluate all growth funds in exactly the same tournament, the paper reruns the test inside volatility-based subtournaments. The result remains. The paper also notes that Brown and Goetzmann's criticism of fund objective categories does not overturn the main finding.

Interpretation. Even if the true investor comparison set is narrower than "all growth funds," the basic loser-high-risk pattern is still there.

4.8 What the paper can and cannot distinguish

The paper is persuasive on *whether* losers adjust risk more than winners, but not on *how* they do it.

- The data do not observe portfolio holdings, so the paper cannot say whether managers increase risk through turnover, derivatives, leverage, or cash allocation.
- The paper also cannot directly observe investor flows at the fund-month level within the test design, so the convex-payoff mechanism is argued from industry evidence rather than estimated inside the paper.
- Finally, the analysis is limited to growth funds, so external validity to other fund categories is suggestive rather than proven.

5 Tables

5.1 Table I: descriptive statistics for growth funds, 1976–1991

Read:

- The number of funds in the annual sample rises from 109 in 1976 to 334 in 1991.
- Total assets under management increase from \$8.93 billion to \$134.25 billion.
- Median annual returns range from −4.89% in 1990 to 35.32% in 1980.
- Median annual standard deviation also varies widely, from roughly 10.8% to more than 31%.

Economic message. Table I does two jobs. First, it shows that the market environment varies enormously from year to year, so relative rankings are economically meaningful. Second, it documents the explosive growth of the growth-fund segment, which motivates the later claim that tournament incentives became stronger as the industry expanded.

5.2 Table II: the whole-sample baseline

Read:

- Table II reports the core 2x2 contingency-table evidence for the full 1980–1991 sample.

Table I

Descriptive Statistics for a Sample of 334 Growth-Oriented Mutual Funds, 1976–1991

Summary information is reported for the sample of mutual funds with a growth orientation, as maintained by Morningstar Mutual Fund Services. A fund is only included in an annual sample if it has return data for the entire year.

Year	Number of Funds	Total Dollar Investment[a]	Median Return[b]	Median Std. Deviation[b]
1976	109	$ 8.93	21.97%	16.07%
1977	112	9.11	1.75	11.60
1978	118	10.71	12.12	21.27
1979	120	14.12	30.00	16.70
1980	121	13.99	35.32	21.48
1981	126	16.97	−1.56	16.07
1982	132	27.89	27.61	19.61
1983	142	29.62	21.11	12.54
1984	156	40.40	−2.29	15.87
1985	174	54.16	29.12	13.65
1986	201	60.01	14.50	17.04
1987	234	65.51	2.67	31.38
1988	266	83.59	14.86	10.81
1989	287	83.23	26.71	11.95
1990	311	129.24	−4.89	19.23
1991	334	134.25	34.31	17.08

[a] Total year-end dollar value of all funds classified as growth oriented, expressed in billions.
[b] Statistics are reported for the median values of the annual fund samples.

Table II

Frequency Distributions of a 2 × 2 Classification of the Risk Adjustment Ratio and "Winner/Loser" Variables, 1980–1991

Cell frequencies are reported for a 2 × 2 classification scheme involving the rank-ordered variables: (i) the Risk Adjustment Ratio (*RAR*); and (ii) the compound total return through the first *M* months of the year (*RTN*). We employ five different values for the interim performance assessment date: $M = 4$, 5, 6, 7, and 8. *RAR* is the ratio of standard deviations measured before and after month *M*. We construct and normalize the data for the classifications on a yearly basis from monthly returns to 334 growth-oriented mutual funds and aggregated for the 1980–1991 sample period. We divide the funds into four groups on a yearly basis according to (i) whether *RTN* is below ("low" or "loser") or above ("high" or "winner") the median (Panel A) *or* in the highest ("winner") or lowest ("loser") quartile (Panel B), and (ii) whether *RAR* is above ("high") or below ("low") the median. We also list results for *RAR* rankings calculated both with and without December returns.

		Sample Frequency (% of Observations)					
		Low *RTN* ("Losers")		High *RTN* ("Winners")			
Assessment Period[a]	Observations	"Low" *RAR*	"High" *RAR*	"Low" *RAR*	"High" *RAR*	χ^2	*p*-value[b]
Panel A: Winners/Losers Ranked by Median							
A1: All Monthly Returns							
(4, 8)	2484	24.32	25.60	25.56	24.52	1.36	0.244
(5, 7)		22.83	27.09	27.09	22.99	17.42	0.000
(6, 6)		22.46	27.46	27.46	22.62	23.97	0.000
(7, 5)		22.14	27.78	27.70	22.38	29.79	0.000
(8, 4)		23.43	26.49	26.45	23.63	8.58	0.003
A2: Excluding December Returns							
(4, 8)	2484	24.52	25.40	25.36	24.72	0.58	0.446
(5, 7)		23.27	26.65	26.61	23.47	10.57	0.001
(6, 6)		22.79	27.13	27.13	22.95	18.10	0.000
(7, 5)		22.22	27.70	27.66	22.42	28.49	0.000
(8, 4)		23.39	26.53	26.53	23.55	9.30	0.002
Panel B: Winners/Losers Ranked by Quartile							
B1: All Monthly Returns							
(4, 8)	1233	25.61	24.31	26.01	24.07	0.05	0.821
(5, 7)		23.20	26.68	28.95	21.17	15.67	0.000
(6, 6)		22.63	27.25	28.63	21.49	17.05	0.000
(7, 5)		21.65	28.22	27.41	22.71	15.69	0.000
(8, 4)		22.71	27.17	28.06	22.06	13.49	0.000
B2: Excluding December Returns							
(4, 8)	1233	25.69	24.23	26.18	23.91	0.08	0.777
(5, 7)		23.20	26.68	28.39	21.74	12.67	0.000
(6, 6)		22.47	27.41	28.47	21.65	17.05	0.000
(7, 5)		21.57	28.30	27.58	22.55	17.07	0.000
(8, 4)		21.98	27.90	27.90	22.22	16.59	0.000

[a] The performance assessment period is listed as (*M*, 12-*M*), where *M* indicates the month of the interim assessment and 12-*M* is the rest of the year.
[b] The χ^2 statistic is calculated based on a null hypothesis that each cell should receive an equal distribution (i.e., 25 percent) of the sample.

- For the July assessment date ($M = 7$) with December excluded, the loser/high-risk-adjustment and winner/low-risk-adjustment cells are 27.70% and 27.66%, versus 22.22% and 22.42% for the opposite diagonal.
- The corresponding chi-square statistic is 28.49, with a p-value essentially equal to zero.
- Results are weak for the April cutoff ($M = 4$), but strong from May through August, with July giving the sharpest separation.
- Median-based and quartile-based winner-loser classifications tell very similar stories.

Economic message. This is the paper's main stylized fact. The pattern is exactly what the tournament hypothesis predicts: losers are too often found in the high-risk-adjustment cell, and winners are too often found in the low-risk-adjustment cell. The fact that the result is strongest around the July ranking date also fits the idea that second-quarter rankings are especially salient to investors and the financial press.

5.3 Table III: temporal dynamics

Read:

- Table III fixes the preferred specification (median ranking, $M = 7$, December excluded) and splits the sample over time.
- For the whole period 1980–1991, loser/high-RAR and winner/low-RAR each occur about 27.7%,

with a chi-square of 28.49.

- For 1980–1985, the chi-square is only 1.98, so the winner-loser difference is not statistically important.
- For 1986–1991, the chi-square jumps to 57.72, and the key cells rise to 29.70% each.
- In the last three-year subperiod, 1989–1991, the loser/high-RAR and winner/low-RAR cells each reach 31.22%, versus only 18.78% for the opposite diagonal.

Economic message. Table III says the effect is not a timeless constant. It becomes much stronger late in the sample, exactly when investor awareness, industry size, and media attention to performance rankings increased sharply.

Table III

Temporal Partitions of the Risk Adjustment Ratio and "Winner/Loser" Variable Classifications Using Median Values: December Observations Excluded

Listed below are the cell frequencies for a 2 × 2 classification scheme involving the rank-ordered variables: (i) the Risk Adjustment Ratio (*RAR*); and (ii) the total return (*RTN*) through the first seven months (i.e., $M = 7$) of the year. We construct and normalize the data for the classifications on a yearly basis from monthly returns to 334 growth-oriented mutual funds during the period 1980 to 1991. We divide the funds into four groups on a yearly basis according to whether *RTN* is below ("low" or "loser") or above ("high" or "winner") the median and whether *RAR* is above ("high") or below ("low") the median. Results are reported for four sample partitions: (i) the whole sample, (ii) two six-year periods, (iii) four three-year periods, and (iv) twelve annual periods. *RAR* values are calculated and ranked excluding December observations for each sample year.

		Sample Frequency (% of Observations)					
		Low *RTN* ("Losers")		High *RTN* ("Winners")			
Sample Period	Observations	"Low" *RAR*	"High" *RAR*	"Low" *RAR*	"High" *RAR*	χ^2	*p*-value[b]
Panel A: Whole Sample							
1980–1991	2484	22.22	27.70	27.66	22.42	28.49	0.000
Panel B: Six-Year Periods							
1980–1985	851	26.09	23.85	23.74	26.32	1.98	0.160
1986–1991	1633	20.21	29.70	29.70	20.39	57.72	0.000
Panel C: Three-Year Periods							
1980–1982	379	27.18	22.69	22.43	27.70	3.61	0.057
1983–1985	472	25.21	24.79	24.79	25.21	0.03	0.854
1986–1988	701	22.11	27.82	27.82	22.25	8.90	0.003
1989–1991	932	18.78	31.22	31.22	18.78	55.78	0.000

[a] The χ^2 statistic is calculated based on a null hypothesis that each cell should receive an equal distribution (i.e., 25 percent) of the sample.

Table IV

Comparative Frequency Distributions for New versus Entrenched Funds and for Small versus Large Funds

Listed below are the cell frequencies for a 2 × 2 classification scheme involving the rank-ordered variables: (i) the Risk Adjustment Ratio (*RAR*); and (ii) the total return (*RTN*) through the first seven months (i.e., $M = 7$) of the year. We divide the funds into four groups on a yearly basis according to whether *RTN* is below ("loser") or above ("winner") the median and whether *RAR* is above ("high") or below ("low") the median. Entrenched funds are those that were in existence for the whole sample period while "new" funds enter during the period. We define "small" ("large") funds as having a net asset value below (above) the median for a particular tournament year. Results are listed for both the entire 1980–1991 period and the two six-year subsamples. *RAR* values are calculated and ranked excluding December observations for each sample year.

		Sample Frequency (% of Observations)					
		Low *RTN* ("Losers")		High *RTN* ("Winners")			
Period	Sample (No. of Obs.)	"Low" *RAR*	"High" *RAR*	"Low" *RAR*	"High" *RAR*	χ^2	*p*-value[b]
Panel A: New vs. Entrenched Funds							
1980–91	Entrenched ($n = 1341$)	23.56	25.80	26.92	23.71	3.96	0.047
	New ($n = 1143$)	20.65	29.92	28.52	20.91	32.53	0.000
1980–85	Entrenched ($n = 669$)	26.31	24.22	22.57	26.91	2.79	0.095
	New ($n = 182$)	25.27	22.53	28.02	24.18	0.01	0.913
1986–91	Entrenched ($n = 672$)	20.83	27.38	31.25	20.54	19.74	0.000
	New ($n = 961$)	19.77	31.32	28.62	20.29	37.75	0.000
Panel B: Small vs. Large Funds							
1980–91	Large ($n = 1382$)	20.26	24.82	29.59	25.33	10.95	0.001
	Small ($n = 1102$)	24.68	31.31	25.23	18.78	19.03	0.000
1980–85	Large ($n = 482$)	24.69	22.82	23.24	29.25	2.85	0.091
	Small ($n = 369$)	27.91	25.20	24.39	22.49	0.01	0.919
1986–91	Large ($n = 900$)	17.89	25.89	33.00	23.22	28.19	0.000
	Small ($n = 733$)	23.06	34.38	25.65	16.92	29.02	0.000

[a] The χ^2 statistic is calculated based on a null hypothesis that each cell should receive an equal distribution (i.e., 25 percent) of the sample.

5.4 Table IV: new versus entrenched, small versus large

Read:

- Panel A compares entrenched funds to newer funds.
- Over 1980–1991, new funds show a much sharper pattern: loser/high-RAR is 29.92% and winner/low-RAR is 28.52%, with a chi-square of 32.53. Entrenched funds show a weaker effect, with 25.80% and 26.92% in the key cells and a chi-square of 3.96.
- Panel B compares small and large funds. The full-sample pattern is stronger for small funds: loser/high-RAR is 31.31% and winner/high-RAR only 18.78%, versus 24.82% and 25.33% for large funds.
- Both dimensions become much stronger in 1986–1991.

Economic message. Table IV supports the secondary hypothesis. Tournament effects are present for both types of funds, but they are stronger where one would most expect aggressive behavior: among newer and smaller funds that have less to protect and more to gain from a turnaround.

5.5 Table V: current and cumulative performance

Read:

- Table V estimates logistic regressions that relate risk-adjustment outcomes to current interim performance and to two- and four-year cumulative performance.
- In Panel A, the coefficient on the current winner dummy is 0.430; adding a two-year winner indicator gives 0.365 on current performance and 0.225 on two-year cumulative performance; adding a four-year measure gives 0.390 and 0.367.
- In Panel B, where performance is measured in levels rather than ranked dummies, the coefficients remain positive and significant.
- The paper interprets these numbers as implying that an interim loser has a higher probability of being in the high-risk-adjustment group than an interim winner.

Economic message. Table V says the tournament effect is not just about one bad half-year. Managers with a history of poor relative performance are more prone to aggressive post-assessment risk changes. In other words, persistent loser status intensifies the incentive to gamble.

Table V

The Relationship Between Risk Adjustments and Past Performance

Listed below are the estimated coefficients for various forms of the following logistic regression: *DVOLRAT* = f(Interim Annual Return, 2-year Cumulative Return, 4-year Cumulative Return), where *DVOLRAT* is defined as a binary variable assuming the value of 1 if a fund's volatility ratio is below the median value in an annual tournament, 0 otherwise. We use two sets of independent variable definitions. In Panel A, they are defined by assigning a value of 1 to any fund whose return is above the median in the interim annual, 2-year cumulative, or 4-year cumulative assessment periods (i.e., "winners"), 0 otherwise. Panel B then reports results using actual values for the independent variables.[a]

No. of Obs.	Intercept	Interim Perform. Measure	Cum. 2-yr Perform. Measure	Cum. 4-yr Perform. Measure	χ^2
		Panel A: Binary Variable (i.e., Ranked) Values			
2484	−0.220 (0.000)	0.430 (0.000)			28.49 (0.000)
2079	−0.336 (0.000)	0.365 (0.000)	0.225 (0.011)		24.40 (0.000)
1752	−0.391 (0.000)	0.390 (0.000)		0.367 (0.000)	32.27 (0.000)
		Panel B: Actual Variable Values			
2484	−0.005 (0.909)	0.200 (0.000)			23.37 (0.000)
2079	−0.039 (0.378)	0.149 (0.001)	0.118 (0.008)		18.73 (0.000)
1752	−0.009 (0.846)	0.163 (0.002)		0.136 (0.006)	20.01 (0.000)

[a] *P*-values for the Wald chi-squared statistics associated with each regression coefficient as well as for the covariates are listed parenthetically beside each statistic.

Table VI

Risk Adjustment Ratio Classifications for an Unmanaged Control Portfolio Sample

Listed below are the cell frequencies for a 2 × 2 classification scheme involving the rank-ordered variables: (i) the Risk Adjustment Ratio (*RAR*); and (ii) the total return (*RTN*) through the first seven months (i.e., $M = 7$) of the year. We construct and normalize the data for the classifications on a yearly basis from monthly returns to a set of 250 simulated portfolios. Each control portfolio contains stocks randomly selected during the period 1980 to 1991 and the composition of a fund is not altered once formed. We divide funds into four groups on a yearly basis according to whether *RTN* is below ("low" or "loser") or above ("high" or "winner") the median and whether *RAR* is above ("high") or below ("low") the median. Results are reported for three sample partitions: (i) the whole sample, (ii) the last six-year period, and (iii) the last two three-year periods. Differences in cell values relative to the actual fund sample are also reported.

		Control Sample Frequency (% of Observations)					
		Low *RTN* ("Losers")		High *RTN* ("Winners")			
Sample Period	Observations	"Low" *RAR*	"High" *RAR*	"Low" *RAR*	"High" *RAR*	χ^2	*p*-value[a]
		Panel A: 75-Stock Control Portfolios					
1980–1991	3000	23.97	26.00	26.00	24.03	13.72	0.000
	Difference[b]	−1.75	1.70	1.66	−1.61		
1986–1991	1500	25.00	25.00	25.00	25.00	54.98	0.000
	Difference	−4.79	4.70	4.70	−4.61		
1986–1988	750	27.60	22.40	22.40	27.60	35.71	0.000
	Difference	−5.49	5.42	5.42	−5.35		
1989–1991	750	22.40	27.60	27.60	22.40	16.76	0.000
	Difference	−3.62	3.62	3.62	−3.62		
		Panel B: 150-Stock Control Portfolios					
1980–1991	3000	23.73	26.17	26.17	23.93	11.12	0.001
	Difference	−1.51	1.53	1.49	−1.51		
1986–1991	1500	25.07	24.87	24.87	25.20	58.12	0.000
	Difference	−4.86	4.83	4.83	−4.81		
1986–1988	750	25.87	24.13	24.00	26.00	17.13	0.000
	Difference	−3.76	3.69	3.82	−3.75		
1989–1991	750	24.27	25.60	25.73	24.40	39.46	0.000
	Difference	−5.49	5.62	5.49	−5.62		

[a] The χ^2 statistic is calculated based on a null hypothesis that each cell should receive a distribution equal to that of the actual sample (cf. Table III).
[b] Indicates the difference in respective cell frequencies between actual and control fund samples, calculated as the former minus the latter.

5.6 Table VI: unmanaged control portfolios

Read:

- Table VI compares actual mutual funds to simulated unmanaged stock portfolios containing either 75 or 150 stocks.
- In both control samples, the actual-fund frequencies are more skewed in the tournament-predicted direction.

Table VII

Comparative Frequency Distributions for Volatility-Ranked Subtournaments, 1986–1991

Listed below are the cell frequencies for a 2 × 2 classification scheme involving the rank-ordered variables: (i) the Risk Adjustment Ratio (*RAR*); and (ii) the total return (*RTN*) through the first seven months (i.e., $M = 7$) of the year. We initially divide funds into four groups on a yearly basis according to whether *RTN* is below ("loser") or above ("winner") the median and whether *RAR* is above ("high") or below ("low") the median. We then further sort the funds into one of three non-overlapping "volatility" subtournaments at the beginning of each year according to the relative level of their return volatility from the previous two years. We use two different measures of volatility: (i) total risk, measured by return standard deviation; and (ii) systematic risk, measured by an individual fund's beta coefficient relative to an equally weighted average of the entire sample. Results are listed for the six-year period 1986–1991. *RAR* values are calculated and ranked excluding December observations for each sample year.

		Sample Frequency (% of Observations)					
		Low *RTN* ("Losers")		High *RTN* ("Winners")			
Volatility Subgroup	No. of Obs.	"Low" *RAR*	"High" *RAR*	"Low" *RAR*	"High" *RAR*	χ^2	*p*-value[b]
Panel A: Total Risk							
High	492	20.93	28.86	28.86	21.34	11.74	0.001
Middle	492	21.54	28.25	28.25	21.95	8.33	0.004
Low	489	20.25	29.45	29.45	20.86	15.49	0.000
Panel B: Systematic Risk							
High	492	20.93	28.86	28.86	21.34	11.74	0.001
Middle	492	20.73	29.07	29.07	21.14	13.01	0.000
Low	489	22.09	27.61	27.61	22.70	5.33	0.021

[a] The χ^2 statistic is calculated based on a null hypothesis that each cell should receive an equal distribution (i.e., 25 percent) of the sample.

- For example, in the 75-stock control sample for 1986–1991, the actual-minus-control differences are −4.79, +4.70, +4.70, and −4.61 across the four cells, meaning actual funds show more loser/high-RAR and winner/low-RAR mass than unmanaged portfolios do.
- The same qualitative result appears for 150-stock controls.

Economic message. Table VI is the paper's key active-management check. If losers looked riskier later in the year only because loser portfolios mechanically became more volatile, unmanaged control portfolios should look similar. They do not. That strongly suggests deliberate managerial action.

5.7 Table VII: volatility subtournaments

Read:

- Table VII repeats the winner-loser test within prior-volatility subgroups over 1986–1991.
- Using prior two-year total risk, the loser/high-RAR and winner/low-RAR cells remain around 28%–29% in all three subgroups, with significant chi-square statistics.
- Using beta-based subgroups, the result also survives: high-beta, middle-beta, and low-beta groups all show the predicted pattern, with p-values from 0.001 to 0.021.

Economic message. Table VII shows that the core effect is not driven by objective-style mixing or by different responses to aggregate volatility across high- and low-beta funds. Even inside relatively homogeneous volatility groups, losers still adjust risk more than winners.

6 Short summary

- The paper models the mutual fund industry as a tournament in which compensation depends on relative performance because winning funds attract more new money.

- It predicts that managers who are behind at an interim assessment date should take more risk over the rest of the year than managers who are ahead.
- Using 334 growth-oriented mutual funds, the paper finds exactly this pattern: interim losers are more likely to fall into the high risk-adjustment group, especially when rankings are taken around July.
- The effect becomes much stronger in the late 1980s and early 1990s, when the industry grows rapidly and investor attention intensifies.
- Newer and smaller funds display stronger effects, and repeated loser status reinforces the tendency to raise risk.
- Simulated unmanaged portfolios and ex ante volatility subgroup tests suggest that the pattern reflects active managerial behavior rather than passive changes in stock volatility or simple market-risk differences.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the relative-performance tournament structure of the mutual fund industry appears to affect managerial behavior in a systematic way. Managers who are behind at a midyear ranking raise portfolio risk more than managers who are ahead, and this tendency is strongest when fund competition and investor attention are most intense.

7.2 Economic intuition

The core intuition is simple. If compensation is convex in rank because winners attract large inflows and losers are not punished symmetrically, then a manager with poor interim performance has an incentive to gamble. The paper's evidence says that this is not just a theory: loser funds do, on average, move toward relatively higher second-half risk, and they do so in a way that looks deliberate rather than mechanical.

7.3 Takeaway

The deepest message of the paper is not merely that managers chase risk. It is that a market institution that looks harmless—annual performance rankings combined with flow-sensitive compensation—can distort portfolio policy. In that sense, the paper is an early and influential warning that tournaments may discipline managers in some settings but may also push them toward short-horizon behavior that investors never explicitly asked for.